

BEOMSEOK SEO

Ph.D., Assistant Professor, Dept. of Statistics, Sookmyung Women's University

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EDUCATION

Ph.D., Statistics, **Penn State University**, University Park, PA 2016 - 2021

- Thesis: [Interpretable Statistical Learning: From Hidden Markov Models to Neural Networks.](#)

B.A., Economics & Statistics, **Korea University**, Seoul, Korea 2005 - 2011

EMPLOYMENT

Assistant Professor, **Sookmyung Women's Univ.**, Dept. of Stat., Seoul, Korea 2024 - Present

Adjunct Professor, **Sungkyunkwan Univ.**, Dept. of Quant. Applied Econ., Seoul, Korea
2023 - Present

Adjunct Professor, **Korea Univ.**, Dept. of Stat., Seoul, Korea 2023 - 2024

Senior Economist, **The Bank of Korea**, Seoul, Korea 2011 - 2024

- Research Dept., Office of Econ. Modeling & Policy Analysis, Economic Statistics Dept.,
Financial Markets Dept.

RESEARCH INTERESTS

My research interests include a range of related problems in

- Interpretable statistical modeling and machine learning,
- Natural language processing and text mining,
- Development of statistical tools for addressing economic and financial challenges.

PUBLICATION

* Corresponding author

Beomseok Seo*, Hyungbae Cho, Dongjae Lee (2026). ["Point and Risk estimation Using an ensemble of Models for Nowcasting: PRISM-Now."](#) Journal of Forecasting, 45(6), 2760-2784.

- Yu-Ling Chang, **Beomseok Seo*** (2025). ["Using Explainable Artificial Intelligence to Envision the Future of Work and the Workforce – From Prediction to Explanation."](#) Technology in Society, 103139.
- Jaewon You, **Beomseok Seo***, Junyoung An (2025) [" Who Pays When Zombie Firms Persist? Asymmetric Debt Conditions by Credit Rating."](#) Finance Research Letters, 108616.
- Jaewon You, **Beomseok Seo*** (2025) ["Deposit Competition and Depository Institution Risk: Do Interest Margins Matter?"](#) Emerging Markets Finance and Trade, 1-21.
- Beomseok Seo*** (2025) ["Econometric Forecasting Using Ubiquitous News Text: Text-enhanced Factor Model."](#) International Journal of Forecasting, 41(3), 1055-1072.
- Beomseok Seo***, Jia Li (2024) ["Explainable machine learning by SEE-Net: closing the gap between interpretable models and DNNs."](#) Scientific Reports, 14, 26302.
- Beomseok Seo***, Younghwan Lee, Hyungbae Cho (2024) ["Measuring News Sentiment of Korea Using Transformer."](#) Korean Economic Review (SSCI), 40(1), 149-176.
- Beomseok Seo***, Lynn Lin, Jia Li (2022) ["Mixture of Linear Models Co-supervised by Deep Neural Networks."](#) Journal of Computational and Graphical Statistics, 31(4), 1303-1317.
- Beomseok Seo***, Lynn Lin, Jia Li (2021). ["Block-wise Variable Selection for Clustering via Latent States of Mixture Models."](#) Journal of Computational and Graphical Statistics, 31(1), 138-150.
- Jia Li, **Beomseok Seo**, Lynn Lin (2019) ["Optimal Transport, Mean Partition, and Uncertainty Assessment in Cluster Analysis."](#) Statistical Analysis and Data Mining: The ASA Data Science Journal, 12(5), 359-377.
- Seung Jun Shin, **Beomseok Seo***, (2024) ["Real-Time Private Consumption Prediction Using Big Data."](#) (in Korean). Korean Journal of Applied Statistics, 37(1), 13-38.
- Jae Keun Yoo, Yujin Park, **Beomseok Seo***, (2024) ["Using Noise Filtering and Sufficient Dimension Reduction Method on Unstructured Economic Data."](#) (in Korean). Korean Journal of Applied Statistics, 37(2), 119-138.

WORK IN PROGRESS

- Young Joon Lee, Soohyon Kim, **Beomseok Seo**, Ki Young Park "Measuring Monetary Policy Surprises Using Text Mining: The Case of Korea." R&R
- Beomseok Seo**, Jaewon You, Yucheol Noh "When Does Housing Policy Stabilize the Housing-Credit Cycle? Policy Endogeneity and Text-Based Evidence from Korea" Submitted

Beomseok Seo*, Habin Choi "LINC-Split: Linearly Interpretable Neural Clustering via Local Splits." Will be submitted soon

TEACHING

Statistics & Engineering

- Statistical Methods I (MA in Stat SMU)
- Text Mining and Natural Language Processing (UG in Stat SMU)
- Statistical Machine Learning (UG in Stat SMU)
- Practicum in Statistical Analysis (UG in Stat SMU)
- Introduction to Probability and Stochastic Processes (UG in Engineering PSU)
- Elementary Statistics (UG in Stat PSU)

Economics

- Big Data Analytics in Macroeconomics (MA in Econ SKKU)
- Machine Learning for Finance and Economics (MA in Stat KU)
- Topics in Data Information (MA in Policy Studies KU)
- (Speical topic) AI and Economic Forecasting (Working Professionals)

SELECTED CONFERENCE & SEMINAR TALKS

<i>Mirae-Asset Management: Innovative Growth Forum, Seoul</i>	<i>July 2026</i>
<i>The Korean Statistical Society Conference (TKSS), Gyeongju</i>	<i>July 2026</i>
<i>Korea Housing Finance Corporation (HF): AI and Economic Forecasting, Busan</i>	<i>Jun 2026</i>
<i>Mirae-Asset Management: Strategic Application of AI in Financial Data, Seoul</i>	<i>Apr 2026</i>
<i>Toss insight: Text mining in Economic Analysis, Seoul</i>	<i>Mar 2026</i>
<i>Mirae-Asset Management: AI in Economic Analysis, Seoul</i>	<i>Dec 2025</i>
<i>International Forum on AI Risks & AI Ethics Review at NTNU (keynote speech), Taipei</i>	<i>Dec 2025</i>
<i>The Korean Statistical Society (TKSS) & Bank of Korea (BOK) Joint Forum, Seoul</i>	<i>Sep 2025</i>
<i>Bank of the Lao PDR (BOL): Text-mining for Financial Literacy, Vientien</i>	<i>July 2025</i>
<i>Office of Economic Modeling and Policy Analysis at BOK: AI in Economic Analysis, Seoul</i>	<i>July 2025</i>
<i>Korea Data Analysis Society (KDAS) conference: Econ. Forecasting Using Text, Seoul</i>	<i>July 2025</i>
<i>Korea Deposit Insurance Corporation (KDIC): Machine Learning for Econ. Prediction, Seoul</i>	<i>Mar 2025</i>

<i>Sungkyunkwan Univ. Dept of Econ., Seoul</i>	<i>Oct 2024</i>
<i>Korea Development Institute (KDI): Consumption Prediction Using Big-data, Sejong</i>	<i>Jul 2024</i>
<i>International Symposium on Forecasting (ISF): Econ. Forecasting Using Text, Dijon</i>	<i>Jul 2024</i>
<i>Chung-nam institute (CNI): Consumption Prediction Using Big-data, Gongju</i>	<i>Jun 2024</i>
<i>Bank of Korea, BOK Friday Lecture: Machine Learning for Econ. Prediction, Seoul</i>	<i>Feb 2024</i>
<i>Czech National Bank (CNB), Macroeconomic Forecasting Division Seminar, Prague</i>	<i>Oct 2023</i>
<i>European Central Bank (ECB), DG-E Wide Seminar, Frankfurt</i>	<i>Oct 2023</i>
<i>Bank for International Settlements (BIS), IDE and Machine Learning Community, Basel</i>	<i>Oct 2023</i>
<i>The Korean Statistical Society & Bank of Korea Joint Forum, Seoul</i>	<i>Sep 2022</i>
<i>Central Bank Research Association (CEBRA) Conference, Barcelona</i>	<i>Aug 2022</i>
<i>Korea Information Society Development Institute (KISDI), Seoul</i>	<i>Jun 2022</i>
<i>The Korean Statistical Society Conference (TKSS), Seoul</i>	<i>Jun 2022</i>
<i>Bank of Korea Statistics Forum, Seoul</i>	<i>Nov 2021</i>
<i>The Korean Statistical Society Conference (TKSS), Seoul</i>	<i>Nov 2021</i>
<i>Computational and Financial Econometrics (CFE), London</i>	<i>Dec 2019</i>
<i>Joint Statistical Meetings(JSM), Denver</i>	<i>Jul 2019</i>
<i>Joint Statistical Meetings(JSM), Vancouver</i>	<i>Jul 2018</i>

FELLOWSHIPS, HONORS, AND AWARDS

<i>Excellent Faculty Award in Technology Transfer, Sookmyung Women's Univ.</i>	<i>May 2025</i>
<i>Excellent Employee Award for Research, Bank of Korea.</i>	<i>Jun 2023</i>
<i>Teaching Fellowship, Penn State Univ.</i>	<i>Spring 2017 - Spring 2021</i>
<i>Best Presentation Award, Penn State Univ.</i>	<i>May 2018</i>
<i>Graduate Study Fellowships, Bank of Korea.</i>	<i>Fall 2016 - Spring 2018</i>
<i>Best Honors Scholarship, Korea Univ.</i>	<i>Fall 2010</i>
<i>Academic Excellence Scholarship, Korea Univ.</i>	<i>Spring 2006, Fall 2009, Spring 2010</i>

SELECTED FUNDED PROJECTS

<i>Nowcasting Using Time-series Foundational Models Based on LLM Architecture, Bank of Korea (BOK)</i>	<i>2026</i>
<i>Study on Tactical Asset Allocation Strategies Using AI, Mirae-Asset Global Investments</i>	<i>2026</i>

Analysis of Loans for Low-to-Medium Credit Borrowers, Toss	2026
Forecasting Household Debt and the Housing Market with Policy Changes and Market Sentiment, Bank of Korea (BOK)	2025
Study on the Criteria for Detecting Accounting Fraud Risk, Korean Institute of Certified Public Accountants (KICPA)	2025
GDP nowcasting using large-scale corporate and industrial data, Bank of Korea (BOK)	2025
Development of an AI-Based Open Art Platform for Space-Customized Artwork Recommendation and Installation Integration	2025
Forecasting Exchange Rate Volatility Using Big Data, National Assembly Budget Office (NABO)	2025
Early Warning System Construction for Detecting a Bank Run, Korean Deposit Insurance Corporation (KDIC)	2024
Using Noise Filtering and Sufficient Dimension Reduction Method on Unstructured Economic Data, Bank of Korea (BOK)	2022
Real-Time Private Consumption Prediction Using Big Data, Bank of Korea (BOK)	2021

PATENTS

Apparatus for Constructing Economic Indicators Based on News Texts, and Method and System for Business Cycle Forecasting Using Text-Based Indicators	(10-2022-0059258)
Method and apparatus for training and inference of synchronized explainable neural networks	(10-2025-0121763)

PROFESSIONAL EXPERIENCE

<i>National Statistics Committee Member</i> of the Economic Statistics Subcommittee, Ministry of Data and Statistics	2026-2027
<i>Advisory Committee Member</i> of the Office of Economic Modeling and Policy Analysis, Bank of Korea (BOK)	2025-2026
<i>Advisory Committee Member</i> of the Real-Time Economic Diagnosis, Korea Development Institute (KDI)	2024
<i>Advisory Committee Member</i> of the Small Business Market Business Survey Index(BSI), Small Enterprise & Market Service (SEMAS)	2024
<i>Recruitment Committee Member</i> of Korea Real Estate Board (REB)	2026
<i>Recruitment Exam Committee Member</i> of the Korea Securities Finance Corp. (KSFC)	2024
<i>Recruitment Exam Committee Member</i> of the Bank of Korea (BOK)	2021

JOURNAL REVIEW SERVICE

Computational Statistics and Data Analysis, Neural Networks, Journal of Forecasting, Journal of Economic Theory and Econometrics, Etc.

LANGUAGES

English, Korean (fluent)

PROGRAMMING LANGUAGES

Python/Tensorflow, R, SQL (advanced), MATLAB, C/C++, CSS (intermediate)

PROGRAMS

[Text Indices Hub](#), *Interactive website*. (Launching initially takes several minutes.)

- Providing Theme Frequency in News Indices (TFNI), Text-based Business Confidence Indicators (TBCI).

[Tspoon](#), *Python modules on PyPI*,

- Time-series pre-processing, period conversion, normalization, visualization, and more

[MLM](#), *Python modules on GitHub*,

- Interpretable non-linear regression or classification based on "Mixture of Linear Models Co-supervised by Deep Neural Networks"

[HDclustVS](#), *R package on GitHub*,

- A Block-wise Variable Selection Method for High-dimensional Clustering via Latent States of Mixture Models

[OTclust](#), *R package on CRAN*,

- Mean Partition, Uncertainty Assessment, Cluster Validation and Visualization Selection for Cluster